

Attribution, Compression, and Nonseparability in Multi-Source Conformally Flat Initial Data

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Abstract

Multi-source general-relativistic initial data are not generally representable as a linear superposition of independently solved source geometries. Rather than treating the resulting nonadditivity solely as an approximation error, we treat its structured remainder as a measurable object. The measurand throughout is the discrepancy between a solved fused scene and a declared separable or compressed representation of it, within a time-symmetric conformally flat extended-source constraint construction. We report four results. First, a counterfactual ablation-response calculus culminating in a two-factor response law,

$$\frac{S^N}{S^L} \approx \frac{\mu_{\text{eff}}}{\mu} C^{\text{lin}},$$

whose ψ^{-3} source-dressing exponent follows analytically from the constraint and whose second factor is the linearized compensation of the remaining scene. Second, a predictive reconstruction whose prediction arm contains no nonlinear constraint PDE solves – only linear field solves and a low-dimensional nonlinear scalar fixed point – recovering the fused field to sub-percent median fidelity, roughly seventy times better than raw linear superposition on the same measure, while reproducing ledger quantities, trough geometry, and ablation responses. Third, a decomposition of the reconstruction residual into a part representable by reweighting the single-source response basis and a basis-orthogonal part with stable, resolution-reproduced geometry. Fourth, survival of the basis-orthogonal part into the electric-Weyl witness: after optimal basis reweighting, the median remaining tidal-error amplitude is 79–99% of the original reconstruction tidal-error amplitude across all tested scenes. We close by defining the enrichment program under which this remainder is to be systematically attacked.

1 Introduction

The Einstein constraint equations do not superpose. Multi-source initial data do not generally yield physical geometries or curvature fields that are additive combinations of independently solved source geometries; even in constructions where an auxiliary conformal field superposes exactly – Brill–Lindquist data being the canonical example – the resulting metric and curvature need not. This fact is well known, and the research community has developed a substantial literature for coping with it.

A large engineering literature treats the nonadditive remainder as a nuisance. In numerical-relativity initial-data construction, superposed data are a convenient starting guess whose constraint violations are then attenuated, damped, or corrected away: attenuation functions for superposed black-hole and neutron-star data, constraint-propagation studies of “skeleton” data, and superposition remedies for boson-star binaries all belong to a mature engineering tradition whose goal is to minimize the error that superposition introduces [1]. The gravitational-decoupling program characterizes, from the analytic side, special symmetric configurations in which a form of source decoupling can be arranged exactly [2]. At the cosmological scale, the averaging and backreaction literature has long emphasized that spatial averaging and nonlinear evolution do not commute, so that coarse descriptions of an inhomogeneous universe omit structure in a controlled but consequential way [3].

For visualizing what slice-level curvature actually does, the tendex and vortex program established the electric/magnetic Weyl decomposition, its eigenstructure, and its tidal reading as a practical instrument family, from weak field through binary-black-hole simulation [4]. The present work adopts its witness layer directly and checks conventions against it.

This paper inverts the figure and ground. We do not attempt to minimize the nonadditive remainder, decouple it away, or average over it. We isolate it, decompose it, propagate it through a curvature witness, measure what controls it, and attach kill conditions to its name. The claim is deliberately narrow. We do not measure “superposition failure in general relativity” in any global sense; we measure the discrepancy between a solved fused scene and a declared separable or compressed representation of that scene, in a specific, fully declared constraint construction. Within that scope, we show the discrepancy is not noise: it obeys a response law, it can be predicted, its basis-irreducible part has stable geometry, and that geometry survives into the electric part of the Weyl tensor.

The methodological foundation is established in a companion series [5, 6, 7, 8, 9] and is invoked here rather than retaught. Four commitments matter for what follows. Attribution is intervention-based: a source’s contribution is defined by counterfactual removal and re-resolution, never by a claim that any region of the field is owned by a source. The electric Weyl tensor E_{ij} is the primary slice-level tidal witness; scalar summaries are downstream of it and declared. Registration, contraction, and scalarization conventions are part of the evidentiary record, not invisible plumbing. And coarse-graining loss is something to be measured rather than assumed – the information-map framing of GCS III is precisely the bridge this paper supplies with numbers [7].

One sentence of larger motivation, carried with its own falsifier: the program behind this paper asks whether fused gravitational scenes contain reproducible structure that no separable source representation can carry – a question that dies, in this construction, if controlled basis enrichment, profile-family variation, and convention variation eliminate the basis-resistant remainder documented below. Everything in the body of the paper is independent of how that larger question resolves.

The paper proceeds as its argument does: construction (Sec. 2), ablation calculus (Sec. 3), predictive compression (Sec. 4), residual structure and basis projection (Sec. 5), curvature witness (Sec. 6), and interpretation with a falsification program (Sec. 7). Three preregistered hypotheses were rejected in the course of this work; they are displayed in the main text because their failure is part of the evidence.

2 Construction

2.1 Family S

We work with time-symmetric ($K_{ij} = 0$), conformally flat initial data,

$$\gamma_{ij} = \psi^4 \delta_{ij},$$

with an extended-source matter prescription referred to as Family S. The prescribed quantity is the York-scaled conformal density

$$\bar{\rho} \equiv \psi^8 \rho,$$

where ρ is the physical energy density. With this scaling the Hamiltonian constraint reads

$$\nabla^2 \psi = -2\pi \bar{\rho} \psi^{-3}, \quad \psi \rightarrow 1 \text{ at infinity.} \quad (\text{F-S})$$

The definition and the equation travel together: the familiar $\nabla^2 \psi = -2\pi \psi^5 \rho$ form is correct only when ρ denotes physical density, and mixing the two conventions is a category error. A scene consists of N sources, $\bar{\rho} = \sum_s \bar{\rho}_s$, each $\bar{\rho}_s$ a prescribed isotropic profile.

One boundary must be drawn explicitly at the outset. Family S is not Brill–Lindquist data. Brill–Lindquist and related exact-vacuum puncture scenes – where ψ is a sum of poles and the geometry is vacuum everywhere off the punctures – form a neighboring benchmark lane against which parts of this program are checked, but they are not the substrate of the present paper. Everything below concerns extended, regular, spatially localized conformal sources solved through Eq. (F-S). The distinction matters because several central objects – effective-source coefficients, compensation integrals, and matter-proxy masks – are defined by integrals over extended sources and have no direct puncture analogue.

2.2 Scenes, ladder, and the declared stress rung

All experiments use three source morphologies, each a seed-deterministic roster of $N = 16$ sources with true isotropic Gaussian profiles,

$$\bar{\rho}_s(\mathbf{x}) = \mu_s \pi^{-3/2} R_s^{-3} \exp\left(-\frac{r_s^2}{R_s^2}\right), \quad (1)$$

which are nowhere compactly supported. Every “matter” or “exterior” statement in this paper therefore refers to the declared analysis masks of Sec. 2.4, not to literal support. The morphologies are: **M1**, clustered (sources gathered into a small number of groups); **M2**, filamentary (sources threaded along a quasi-linear corridor); and **M3**, shell (sources arranged on a common spherical shell, maximizing shared interior volume). Rosters are generated from fixed seeds and regenerate exactly; no stored fields are required for reproduction.

Each morphology is solved along a compactness ladder parameterized by a source-amplitude multiplier

$$f \in \{1/3, 2/3, 1, 4/3, 2\}.$$

The rung $f = 2$ is a pre-declared stress rung. It drives the scenes beyond the certified regime of the first-order response calculus ($\psi_{\max} \approx 1.32\text{--}1.35$). The nonlinear solver itself remains numerically clean there – all $f = 2$ solves pass the same algebraic-residual gate as the in-band rungs – so what the rung stresses is the response approximation, not the solver. It is included to test whether, where, and how the reported approximations begin to degrade. Primary resolution is 64^3 , with 96^3 rungs solved per morphology for resolution gates.

2.3 Solver and gates

The nonlinear constraint, Eq. (F-S), is solved by Picard iteration (relaxation $\omega = 0.6$) with a DST-I spectral Poisson kernel, in float64 throughout, with a monopole far-field boundary condition made self-consistent by three outer loops. Algebraic residuals of all accepted solves satisfy

$$\left\| \nabla^2 \psi + 2\pi \bar{\rho} \psi^{-3} \right\| \leq 8.1 \times 10^{-11}$$

on the declared norm. The sweep underlying Sec. 3 comprises 306 nonlinear solves.

Two implementation lanes exist. The results reported here come from a fresh implementation written independently from the canonical equations, as a deliberate cross-implementation reproduction of an earlier lane’s structure; agreement between the lanes is part of the evidence for the screening law in Sec. 3.3. A further cross-wing reproduction run, on a third independently written stack, has been invited and is pending; the reproduction statement is scoped accordingly.

2.4 Declared conventions

Because several downstream quantities are convention-bearing, we fix them once. Writing $\omega \equiv 2 \ln \psi$ so that $\gamma_{ij} = e^{2\omega} \delta_{ij}$, the coordinate spatial Ricci tensor of the conformally flat slice is

$${}^{(3)}R_{ij} = -(\partial_i \partial_j \omega - \partial_i \omega \partial_j \omega) - \delta_{ij} \left(\nabla^2 \omega + |\nabla \omega|^2 \right), \quad (2)$$

which contains both second-derivative and gradient-quadratic terms. The implemented trace-free object is

$${}^{(3)}R_{\langle ij \rangle} \equiv {}^{(3)}R_{ij} - \frac{1}{3} \delta_{ij} \delta^{kl} {}^{(3)}R_{kl}, \quad (3)$$

evaluated with second-order central differences.

For the slice-level Weyl identification we adopt a zero-anisotropic-stress matter completion at rest: $j_i = 0$ and $S_{\langle ij \rangle} = 0$, where S_{ij} is the spatial stress and angle brackets denote the trace-free spatial part. Pressureless dust at rest is one realization, but the identification requires only the vanishing anisotropic-stress condition. In standard $3 + 1$ form, on a $K_{ij} = 0$ slice,

$$E_{ij} = {}^{(3)}R_{\langle ij \rangle} - 4\pi S_{\langle ij \rangle}, \quad (4)$$

up to the adopted Weyl-sign convention. Hence the declared completion gives

$$E_{ij} = {}^{(3)}R_{\langle ij \rangle} \quad (5)$$

exactly on the slice, Gaussian tails included. A derivation sketch and provenance requirement appear in Appendix A.7. The trace sector remains matter-bearing – the Hamiltonian constraint gives ${}^{(3)}R = 16\pi\rho$ – and is exactly what the trace-free projection discards.

We therefore refer to the implemented object as the electric-Weyl witness for the declared matter completion. On the vacuum benchmark lane the identification reduces to the standard vacuum identity. Tensor amplitudes are compared through the Frobenius norm taken with the flat δ metric; this contraction is a declared chart-level convention, not an invariant.

Two analysis masks are declared once and used everywhere. The *matter-proxy mask* is the union of balls $r_s \leq 2.2R_s$ about each source center – an operational proxy for source-associated support, not a support boundary. The *exterior mask* is its complement with a 10% skin ($r_s > 2.42R_s$ for every s), minus a three-cell frame at the box boundary. On the exterior mask the matter density

is at Gaussian-tail level but not literally zero; the Weyl identification above does not require a small-tail approximation because it is tied to the declared stress completion.

The projection inner product used in Sec. 5.3 is the unweighted L^2 product on the exterior mask, with basis

$$V_1 = \text{span}\{u_{L,s}\} \cup \{1\},$$

where $u_{L,s}$ is the single-source linear response of source s and the constant absorbs monopole offset. All headline quantities are dimensionless ratios, positions, or morphological statistics under these declared conventions; absolute-amplitude comparisons across conventions are deferred to a separate convention audit.

2.5 Preregistration

The primary experimental campaigns and their directional and structural hypotheses were preregistered before their grading computations. Hypotheses, statistics, thresholds, and failure outcomes were filed in advance and are reproduced verbatim in Appendix C with a concordance mapping each registered item to its disposition. Post-hoc analyses – including the initial V_1 residual decomposition of Sec. 5.3 – are explicitly labeled as such; gate experiments subsequently motivated by those analyses were separately preregistered before compute.

One procedural extension is likewise on the record. The surrogate test of Sec. 5.1 was preregistered and originally graded with a 20-realization ensemble, which determined the verdicts. For publication, the identical frozen procedure was rerun with 1,000 realizations to improve empirical- p resolution under a pre-filed declaration, and no verdict changed. Opposite-tail significance reported below is a post-hoc characterization layered on the preregistered rejection, not a retroactive directional hypothesis. Three preregistered hypotheses failed and are reported in the main text with the same prominence as the passes.

3 The ablation calculus

3.1 Four objects, distinguished before anything is measured

The word “attribution” invites a category error, and this section exists partly to make that error impossible. Four objects must be held apart.

- (i) The constraint source term, $\bar{\rho}_s$: the prescribed conformal density contributed by source s to the right-hand side of Eq. (F-S). This is input, not response.
- (ii) The counterfactual ablation response, $\delta\psi_s$: solve Eq. (F-S) with $\bar{\rho} \rightarrow \bar{\rho} - \bar{\rho}_s$ to obtain ψ' , and define

$$\delta\psi_s \equiv \psi - \psi'.$$

This is the operational meaning of “source s ’s contribution”: what the scene loses when s is removed and the constraint is re-solved.

- (iii) The effective-source coefficient,

$$\mu_{\text{eff},s} \equiv \int \bar{\rho}_s \psi^{-3} dV,$$

a scalar functional of the fused solution, weighted by source s ’s profile over the computational/asymptotic domain.

- (iv) A source-owned field: a partition of the solved geometry into parcels belonging to individual sources. This object is never defined and never claimed in this paper.

Everything below is built from (i)–(iii). The response law is a decomposition of ablation responses, not a decomposition of spacetime into owned pieces.

3.2 The ADM identity

With $\psi = 1 + u$ and $u \rightarrow M_{\text{ADM}}/(2r)$ at infinity, Gauss’s theorem applied to Eq. (F-S) yields exactly

$$M_{\text{ADM}} = \int \bar{\rho} \psi^{-3} dV = \sum_s \mu_{\text{eff},s}. \quad (\text{ADM})$$

The coefficients are therefore not merely a diagnostic convenience: $\mu_{\text{eff},s}$ is source s ’s exact share of the scene’s ADM ledger under the declared partition of $\bar{\rho}$, and the bare mass

$$\mu_s \equiv \int \bar{\rho}_s dV$$

is the linear-limit ($\psi \rightarrow 1$) value of the same integral. The deepest one-line statement of this section is that the ψ^{-3} weighting is the ADM measure itself. Because $\psi > 1$ wherever matter deepens the scene, $\mu_{\text{eff},s} < \mu_s$ in any nontrivial configuration. Full derivations are collected in Appendix A.

3.3 The screening law

Linearizing Eq. (F-S) about the fused solution for the removal of source s gives

$$\left(\nabla^2 - 6\pi \bar{\rho}_{\text{ctx}} \psi^{-4} \right) \delta\psi = -2\pi \bar{\rho}_s \psi^{-3}, \quad (\text{LIN-ABL})$$

where $\bar{\rho}_{\text{ctx}} \equiv \bar{\rho} - \bar{\rho}_s$ and $\delta\psi \equiv \delta\psi_s = \psi - \psi'$. The sign convention is fixed so that removal lowers the field and $\delta\psi > 0$; the negative source term drives a positive Poisson-type response exactly as the negative source in Eq. (F-S) drives $\psi - 1 > 0$.

Two structural facts follow. First, the effective source of an ablation is $\bar{\rho}_s \psi^{-3}$, not $\bar{\rho}_s$: the analytic dressing exponent -3 lives in the effective source term and is inherited directly from the constraint nonlinearity. Second, the response propagates through a screened operator whose screening density is the context matter.

For a declared response functional S linear in the source at leading order, the ratio of the nonlinear-scene ablation response to its linear-twin counterpart therefore satisfies, at first order,

$$\frac{S^N}{S^L} \approx \frac{\mu_{\text{eff},s}}{\mu_s}. \quad (\text{SCREEN})$$

This screening/source-dressing law holds with zero fitted parameters. Across all fifteen ladder rungs (three morphologies times five compactness values), the correlation between the measured response ratio and μ_{eff}/μ is 0.93–0.97, reproduced across two independent solver implementations. Formal rank: **Established**; scope: the declared Family S construction.

3.4 A displayed negative: the one-factor exponent is not morphology-universal

A natural corollary hypothesis was preregistered and failed. If Eq. (SCREEN) exhausted the response physics, then a direct regression of measured response ratios against mean source depth should recover an effective exponent near the analytic -3 . It does not, and the failure is morphology-structured: the shell family fits -3.46 to -3.55 (marginal pass), the clustered family -3.80 to -4.08 (marginal fail), and the filamentary family approximately -4.9 (unambiguous fail). The hypothesis of a morphology-universal one-factor exponent is rejected.

The mechanism was identified rather than left as anomaly: the fitted exponent absorbs a second, covarying factor – the compensation of Sec. 3.5 – whose correlation with source depth differs by geometry, and is strongest in the filament, where every source sits inside every other source’s near zone. This yields a binding method rule: a fitted effective exponent obtained from the total nonlinear response is not a direct estimator of the analytic source-dressing exponent unless compensation covariance has been separately removed or controlled. The analytic -3 belongs to the effective source $\bar{\rho}_s \psi^{-3}$; it was never the slope of a contaminated regression.

3.5 Compensation

Removing a source and re-solving does not simply subtract that source’s effective mass from the ledger. The remaining scene relaxes, its ψ falls, and every surviving $\mu_{\text{eff},s'}$ rises. We quantify this with a compensation factor C_s : the fraction by which the scene’s ADM response to the ablation is reduced relative to naive removal of $\mu_{\text{eff},s}$. Equivalently, the scene “reclaims” a fraction $1 - C_s$ of the ablated source’s effective mass; the defining integral is given in Appendix A.

Compensation is lawful in the tested family: the median C_s decreases strictly monotonically with compactness in every morphology (clustered $0.884 \rightarrow 0.707$, filamentary $0.853 \rightarrow 0.653$, shell $0.895 \rightarrow 0.725$ across the ladder). Deeper scenes reclaim more of an ablated source. The filament reclaims the most at every rung, consistent with its geometry maximizing overlap between ablated profile weight and screening context. Formal rank: **Observed**; scope: tested family.

3.6 The two-factor response law

The screening ratio and the compensation factor combine into the section’s principal result,

$$\frac{S^N}{S^L} \approx \left(\frac{\mu_{\text{eff}}}{\mu} \right) C^{\text{lin}}, \quad (\text{TWO-FACTOR})$$

where the second factor is the *linearized* compensation C^{lin} , computed from the screened operator of Eq. (LIN-ABL), not the compensation measured from a full nonlinear re-solve. The reason is structural rather than empirical: the source-dressing factor and C^{lin} belong to the same first-order perturbative construction, while the full-solve compensation imports higher-order response into a factorization built to predict the first-order correction.

In band ($f \leq 4/3$), Eq. (TWO-FACTOR) holds with median residual $\leq 1.6\%$ and correlation 0.994 – 0.999 in every morphology. At the declared stress rung $f = 2$, the comparison between candidate second factors becomes probative: the linearized factor predicts responses to 1.2 – 1.7% error, while the full-solve factor degrades to 3.9 – 5.3% . The first-order-consistent object is the better one even at depth. Formal rank: **Reproduced**; scope: tested morphology family. It is a response decomposition within the declared construction, not a universal gravitational response law.

4 Prediction: compressing a fused scene

4.1 The pipeline

Sections 2–3 supply, unexpectedly, the parts list for a predictive compression of the fused scene: a reconstruction whose prediction arm contains zero nonlinear constraint PDE solves. The arm consists of linear field solves plus one low-dimensional nonlinear scalar fixed point; no nonlinear field equation is solved anywhere in prediction.

1. Solve N single-source linear problems,

$$\nabla^2 u_{L,s} = -2\pi \bar{\rho}_s,$$

with sixteen solves per scene.

2. Determine the effective weights $w_s = \mu_{\text{eff},s}/\mu_s$ by an N -dimensional nonlinear scalar fixed point evaluated on the reconstructed field (Appendix A.6): a nonlinear iteration over N numbers, not over a field.
3. Assemble

$$\psi_{\text{rec}} = 1 + \sum_s w_s u_{L,s}.$$

4. Where ablation responses are wanted, compute C^{lin} per source from the screened linearized operator and predict via Eq. (TWO-FACTOR).

Nonlinear constraint solves appear in this section only as grading references: the pipeline’s outputs are compared against full solutions it never used.

4.2 Grading

The reconstruction was graded on nine evaluation scenes (three morphologies times $f \in \{1, 4/3, 2\}$) plus a 96^3 resolution rung, against preregistered bounds. The normalized field error used wherever a field percentage appears is the pointwise ratio

$$\eta_\psi(\mathbf{x}) \equiv \frac{|\psi_{\text{rec}} - \psi_N|}{\psi_N - 1}, \quad (6)$$

summarized by its median and 90th percentile over the interior domain with the three-cell boundary frame excluded. The linear twin is graded on the identical statistic. Table 1 condenses the outcome; the full per-scene table is reserved for Appendix D.

Every preregistered hypothesis in this experiment passed in all nine scenes. The reconstruction costs roughly $4.5\times$ less than the full solve at 64^3 (wall time per scene on identical hardware: 5.3 s versus 24 s; 23 s versus 106 s at 96^3), with the advantage widening with resolution and depth because the prediction arm uses a fixed number of linear-type field solves rather than iterative nonlinear constraint solves.

The compressed model is very good. Precisely because it is very good, its small remaining error becomes scientifically interpretable. Figure F4 in the planned figure set displays the comparison this paragraph rests on, with the raw-linear-twin baseline drawn on the same axes: on the identical normalized field measure, linear superposition errs by 48–100% where the reconstruction errs by roughly one percent. That factor of approximately 70 is what earns permission to treat what remains as signal.

Table 1: Predictive reconstruction grading summary. “In band” denotes $f \leq 4/3$.

Measure (median, in band)	Result	Preregistered bound
Ledger: μ_{eff} recovery	$\leq 0.15\%$	2%
Field: median η_ψ	0.65–1.2%; p90 $\leq 1.4\%$	5%; 10% p90
Same η_ψ , raw linear twin	48–100%	–
Trough census of E field: matched positions	12/12 (8/8 filament) at ≤ 2 cells, every scene	≤ 3 cells
Compensation C^{lin} vs. measured	corr. 0.997–0.999, $\leq 1.6\%$	5%
Ablation response, fully predictive	0.3–1.3%	5%
Resolution rung (96^3)	all statistics stable to third decimal	–

4.3 The stress rung: graceful degradation

At the pre-declared $f = 2$ rung, beyond the certified regime of the first-order calculus, degradation is graceful: field error rises to approximately 1.7–2.1% median and response error to $\leq 1.7\%$. The rung produced no catastrophic breakdown, and – since approximately 2% is not failure by any declared metric – it did not locate the codec’s failure boundary either; that boundary lies beyond the tested ladder.

What the rung does demonstrate is instructive on two counts. First, the fully predictive law with C^{lin} outperforms, at depth, the descriptive law with the measured full-solve compensation. Second, the ladder is constructed to distinguish law failure from departure from a law’s asymptotic validity regime – a distinction treated as binding method for every ladder in this program, even though this particular ladder ran out of rungs before either was reached. Formal rank for the reconstruction: **Reproduced**; qualifier: predictive; scope: tested morphology family.

5 The residual: what compression forgets

Define the reconstruction residual

$$\delta\psi \equiv \psi_N - \psi_{\text{rec}}, \quad (7)$$

the field the codec fails to carry. Section 4 established that $\delta\psi$ is small. This section establishes that it is not noise – and that where it lives is not where we predicted.

5.1 Two preregistered hypotheses about location, both rejected

Before any residual was computed, two directional hypotheses were registered about where compression error would concentrate. Both were natural; both were wrong. Both failures are displayed at full prominence because they guard against reverse-engineering the interpretation of Sec. 7 from the data.

Rejected hypothesis 1: matter localization. The registered claim was that the residual concentrates on matter,

$$\rho_{\text{loc}} \equiv \frac{\langle |\delta\psi| \rangle_{\text{matter}}}{\langle |\delta\psi| \rangle_{\text{exterior}}} > 1, \quad (8)$$

by analogy with an earlier finding that the linear-vs-nonlinear physics residual localizes on context matter. Observed: $\rho_{\text{loc}} = 0.74\text{--}1.06$ across the six primary scenes, with five of six below unity. The preregistered directional test – positive localization – fails; that is the registered verdict.

A post-hoc characterization, obtained by examining the opposite tail of the same frozen surrogate procedure (1,000 realizations; finite-sample convention $p = (k + 1)/(N + 1)$), sharpens the picture: the five below-unity scenes are individually significant in the lower tail ($p = 0.001\text{--}0.005$, against surrogate distributions centered at 1.000 with standard deviations 0.04–0.10), while the sixth (filamentary, $f = 1$; $\rho_{\text{loc}} = 1.056$) is statistically indistinguishable from its ensemble ($p = 0.72$) in either direction. The 96^3 rung reproduces the below-unity significance ($\rho_{\text{loc}} = 0.825$, $p = 0.005$). The residual is, where it deviates from surrogate expectation at all, anti-localized on the matter-proxy mask – a post-hoc finding layered on a preregistered rejection. The canonical lesson is that residual fields inherit their meaning from the comparison that defines them. The physics residual and the compression residual are different objects with different homes; there is no generic doctrine that “nonlinear residuals live on matter.”

Rejected hypothesis 2: contested-attribution affinity. The registered claim was that compression error accumulates preferentially where source custody is contested – where the top-two linear attribution margins are smallest. Observed correlations between $|\delta\psi|$ and inverse margin are approximately zero or negative. The preregistered directional test – positive affinity – fails family-wide; that is the registered verdict.

The post-hoc opposite-tail characterization (same frozen procedure, 1,000 realizations) gives filamentary correlations of -0.31 and -0.34 , each significant at the ensemble resolution floor (lower-tail $p = 0.001$); the shell scene at $f = 1$ is also significantly negative ($p = 0.033$); and the remaining scenes are consistent with zero. Two readings remain admissible and undistinguished: either contested attribution genuinely does not organize this compression error, or the declared margin statistic is an inadequate witness of the relevant contest. The margin diagnostic is not falsified in general; its explanatory power for this residual is.

5.2 Two preregistered hypotheses about structure, both passed emphatically

Depth persistence. The registered claim was that the normalized residual pattern persists as nonlinear depth increases, with pattern correlation between $f = 1$ and $f = 2$ above 0.8. Observed values are 0.9945, 0.9904, and 0.9953 for the clustered, filamentary, and shell morphologies. Nonlinearity deepens the residual far more than it reorganizes it. The forgotten object behaves as a fixed geometric mode, not regime-dependent debris.

Resolution stability. The registered claim was that the residual pattern survives grid refinement. Observed $64^3 \leftrightarrow 96^3$ pattern correlation is 0.9997, with summary statistics reproduced to the third decimal. We use the earned term *resolution-reproduced*; we do not claim continuum convergence in the strict numerical-analysis sense, which would require further refinement levels.

Together, the compression residual is stable, resolution-reproduced geometry that lives somewhere other than where either registered intuition placed it.

A post-hoc regularity. At every sampled exterior datum in every tested scene, $\delta\psi > 0$: the scalar-weight reconstruction systematically underestimates the exterior field. A candidate mechanism exists – the scalar weight $w_s = \mu_{\text{eff},s}/\mu_s$ compresses a spatially varying dressing into one interior-weighted

number, over-suppressing the reconstructed exterior – but it remains a mechanism candidate awaiting its own preregistration, not an established explanation.

5.3 Projection: the residual has two species

Let

$$V_1 = \text{span}\{u_{L,1}, \dots, u_{L,N}, 1\},$$

which contains everything expressible by reweighting the existing single-source linear responses plus a constant. With projection P_V taken in the declared unweighted L^2 inner product on the exterior mask, decompose

$$\delta\psi = \delta\psi_{\text{rw}} + \delta\psi_{\perp}, \quad \delta\psi_{\text{rw}} \equiv P_V \delta\psi, \quad \delta\psi_{\perp} \equiv (I - P_V) \delta\psi. \quad (9)$$

The *reweightable residual* $\delta\psi_{\text{rw}}$ is the portion of compression error that better bookkeeping could remove while remaining inside the separable basis. The *collective residual* $\delta\psi_{\perp}$ is the portion orthogonal to the entire single-source span – the part that no assignment of per-source weights can carry.

Define the basis-orthogonal fraction as the fraction of the residual’s squared L^2 norm lying outside the span,

$$F_{\perp} \equiv \frac{\|\delta\psi_{\perp}\|_2^2}{\|\delta\psi\|_2^2}. \quad (10)$$

The measured values are approximately 21% (clustered), 18% (filamentary), and 62% (shell). In the clustered and filamentary scenes, most of what the codec forgets is correctable bookkeeping. In the shell – the geometry constructed to maximize shared interior volume – most of what it forgets is not expressible in the separable basis at all.

Terminology is used with its qualifications permanently attached. $\delta\psi_{\perp}$ is a *basis-irreducible collective residual*: irreducible relative to this basis, this inner product, this reconstruction family, and this scene representation. A richer basis – pairwise, screened, nonlinear, or learned collective modes – may represent part or all of it; Sec. 7.4 makes that possibility the centerpiece of the falsification program. The interpretation that shared interior volume causes the morphology ordering is a hypothesis, not a result: shared volume was not the preregistered independent variable, and only one profile family has been tested. Formal ranks: stable residual geometry, **Observed** in the tested family from preregistered structural passes; two-species decomposition and morphology ordering, **Observed** post-hoc, promotion candidate.

6 The curvature witness

6.1 Why this section must exist

Everything in Sec. 5 concerns the conformal factor ψ , which is not mere accounting: it is part of the physical spatial metric, $\gamma_{ij} = \psi^4 \delta_{ij}$. But it is a formulation variable, and the discipline inherited here is strict: ψ exposes structure in the construction variables; E_{ij} tests whether that structure survives in a GR-native tidal witness. A conformal-field result borrows curvature language only after a curvature witness has spoken. The question is whether the collective residual survives passage into E_{ij} , or whether it is a ψ -space artifact that differentiation destroys or renders irrelevant.

6.2 The codec is curvature-faithful

Define the codec tidal discrepancy

$$\Delta E^{\text{comp}} \equiv E[\psi_N] - E[\psi_{\text{rec}}]. \quad (11)$$

Here $E[\psi_{\text{rec}}]$ denotes application of the same declared slice-curvature witness map to the reconstructed metric. It is a grading construction and is not asserted to constitute a separately exact solution of the original Family S constraint with the original matter roster. The same convention applies below to $E[\psi_{\text{rec}} + \delta\psi_{\text{rw}}]$. The reference $E[\psi_N]$ is evaluated on the solved Family S scene under the matter completion declared in Sec. 2.4.

A preregistered concern was a differentiation catastrophe: E involves second derivatives, and a sub-percent field error could in principle become a large tidal error. It does not. At $f = 1$, the median pointwise ratio $\|\Delta E^{\text{comp}}\|_F / \|E^N\|_F$ over the exterior mask is 0.67%, 0.48%, and 0.57% across the three morphologies, against a preregistered ceiling of 10%. At the $f = 2$ stress rung it rises only to 1.14–1.78%, consistent with graceful degradation. The reconstruction is not merely field-faithful but, within this family, tidal-witness-faithful. This substantially strengthens Sec. 4: the compression carries the overwhelming majority of the scene’s tidal structure under the declared witness.

6.3 The object and its statistic

The central object is the *collective curvature residual*,

$$\Delta E_{\perp} \equiv E[\psi_N] - E[\psi_{\text{rec}} + \delta\psi_{\text{rw}}], \quad (12)$$

the electric-Weyl discrepancy that remains after the reconstruction has been granted every correction representable within V_1 . The object is explicitly basis-relative; where ambiguity matters we write the fuller form, V_1 -relative collective curvature residual.

Its summary statistic – the statistic, not the object – is

$$G_{\perp} \equiv \frac{\text{median } \|\Delta E_{\perp}\|_F}{\text{median } \|\Delta E^{\text{comp}}\|_F}. \quad (13)$$

Three things $G_{\perp}$ is not must be stated plainly: it is not a variance fraction, not an L^2 energy fraction, and not an orthogonal decomposition in tensor space. The map $\psi \mapsto E[\psi]$ is nonlinear, so orthogonality in ψ space does not transport into an orthogonal decomposition after passage through the witness. The integrated tensor-error fraction that would answer the percentage question directly is defined in Sec. 7.4 and deferred to the follow-up experiment.

6.4 The witness answers

Against a preregistered survival bar of $G_{\perp} > 0.1$, the measured values across all six scenes are

$$G_{\perp} = 0.875, 0.789, 0.990 \quad (f = 1),$$

and

$$G_{\perp} = 0.923, 0.886, 0.964 \quad (f = 2),$$

for the clustered, filamentary, and shell morphologies respectively. In canonical wording: after optimal V_1 -basis reweighting, the median remaining tidal-error amplitude is 79–99% of the original

codec tidal-error amplitude across the tested scenes. The collective residual does not die at differentiation. It has a substantial, reproducible electric-Weyl footprint.

The structure of Sec. 5 survives into the witness. The shell shows the strongest basis resistance at both depths ($0.990 > 0.875 > 0.789$ at $f = 1$; $0.964 > 0.923 > 0.886$ at $f = 2$), and this ordering was preregistered rather than read off after the fact. The depth persistence is preserved: the spatial pattern of $\|\Delta E_\perp\|_F$ correlates at 0.9888, 0.9961, and 0.9939 between $f = 1$ and $f = 2$ per morphology. The collective curvature residual behaves as a persistent morphology that deepens rather than reorganizes.

6.5 Basis resistance is amplified in the tidal witness

The most striking finding is a change of relative importance across the witness gate, stated carefully in the two different currencies the two levels use. At the ψ level, in the declared L^2 inner product, the reweightable species carries the majority of the residual’s squared L^2 norm in two of three morphologies (approximately 79% and 82% in the clustered and filamentary scenes). At curvature level the available statistic is an amplitude ratio, not a norm fraction, and it reads: after optimal V_1 reweighting, the median remaining tidal-error amplitude is 79–99% of the original codec tidal-error amplitude. The two statistics are not commensurable, and we do not convert between them. Whether a genuine integrated-error-fraction inversion exists is exactly what the $T_\perp$ statistic of Sec. 7.4 is defined to decide. What the present measurements establish is that basis-resistant structure, a minority of the ψ residual’s squared L^2 norm in two morphologies, is not correspondingly minor in the tidal witness.

The canonical statement is therefore: in the tested family, passage from the conformal reconstruction residual to the electric-Weyl discrepancy strongly reduces the relative importance of residual structure correctable by single-source V_1 reweighting, leaving a large basis-resistant tidal discrepancy after the available V_1 correction.

A candidate mechanism, stated only as candidate, is that V_1 confines corrections to the span of independently generated conformal response modes. Passage through the nonlinear curvature map can modify cross-terms as those coefficients change, but it cannot supply conformal directions absent from V_1 . The observed persistence of $\Delta E_\perp$ may therefore reflect tidal sensitivity to precisely those missing conformal directions. This mechanism remains unestablished. The scoped statement earned by the data is that the collective component carries tidal geometry novel relative to the declared independently reweighted single-source representation. The unscoped claim – that no individual-source representation whatsoever can express this structure – is not earned; a richer basis may express it.

Formal ranks: codec curvature fidelity, collective-residual survival, morphology ordering, and depth persistence, all **Observed** in the tested family from preregistered passes; basis-resistance amplification across the witness gate, **Observed** post-hoc; curvature novelty relative to V_1 , **Observed** candidate.

7 Interpretation, limits, and the destruction program

7.1 What the representability failure means

Assemble the arc. The fused scene admits a response calculus. That calculus powers a compression so accurate that its error becomes interpretable. The error splits into correctable bookkeeping and a basis-orthogonal remainder with stable geometry. The remainder survives strongly in the GR-native

tidal witness. What the mathematics has cornered is a *representability failure*: a reproducible component of the fused-scene response is orthogonal to the declared separable basis in conformal-field space, and the resulting basis-resistant structure leaves a persistent tidal discrepancy after every correction that basis admits has been granted.

Orthogonality is established in the declared ψ -space inner product; no tensor-space projection has been performed, and no tensor-space orthogonality is claimed. “Tidal nonseparability” below is defined operationally – resistance of the tidal discrepancy to the declared reconstruction class – not as literal orthogonality in E space.

Only now, with the object measured and its scope fenced, do we name the interpretation it suggests. Within this research program, the working name for the nonseparable content of a fused gravitational scene is the *Commons*. The results sharpen that notion from metaphor into a two-level hypothesis.

Level A: conformal nonseparability. A portion of the fused-scene reconstruction residual lies outside the declared separable source basis,

$$\delta\psi_{\perp} = (I - P_V)\delta\psi \neq 0,$$

with reproducible geometry.

Level B: tidal nonseparability. That basis-resistant conformal structure produces a stable, nonvanishing tidal discrepancy: $\Delta E_{\perp}$ persists in the electric-Weyl witness after all V_1 corrections are granted.

The present work establishes Level B within the tested family and declared convention. The consolidated hypothesis – that fused gravitational scenes contain reproducible relational structure which cannot be represented by the declared independently source-resolved basis and whose omission leaves a persistent signature in GR-native tidal geometry – holds the formal rank **Observed**, with the qualifier *candidate architecture*. It is not Established because the basis class tested is narrow. The honest reading is not “the Commons exists,” but “an object with the Commons’ expected signature exists relative to V_1 , and now has to survive attempts to kill it.”

7.2 Limits

The scope fences are load-bearing and are stated with the same prominence as the results. All findings are confined to: the Gaussian profile family (a single profile family; profile universality is untested and is the first scheduled extension); the V_1 single-source basis (pairwise and higher-order bases untested); the declared chart-level E_{ij} convention (sufficient for the survival gate answered here, insufficient for convention independence; invariant and neighboring physical-frame cross-checks are required before stronger descriptive rank); three 16-source morphologies at two resolutions (resolution-reproduced, not continuum-converged); scalar-weight reconstruction (the codec family itself is a declared choice); and time-symmetric slices ($K_{ij} = 0$ – nothing here touches momentum, spin, or evolution). No absolute-amplitude claims across conventions are made.

7.3 The temporal successor

The static results pose their successor question precisely. A time-dependent extension carries the decomposition per slice – $\delta\psi_{\text{rw}}(t)$, $\delta\psi_{\perp}(t)$, $\Delta E_{\perp}(t)$ – as diagnostic channels over evolving data, and asks whether the collective component at time $t + \Delta t$ is *slice-reconstructible* (determined

by the instantaneous source roster and geometry alone) or *history-bearing* (containing structure recoverable only from the evolved state). The crisp test is whether $\Delta E_{\perp}(t + \Delta t)$ is predicted by the reconstruction operator applied to the instantaneous scene, or measurably is not. No temporal claim of any kind is made in this paper.

7.4 The destruction program

The paper ends not with the interpretation but with the plan to attack it. The collective curvature residual is a basis-relative object, and the correct response to a basis-relative object is basis enrichment. Define nested spaces

$$V_1 = \text{span}\{u_{L,s}\} \cup \{1\} \subset V_2 = V_1 + \text{span}\{u_{st}^{(2)}\} \subset V_3 = V_2 + \{\text{screened/higher-order collective modes}\}, \quad (14)$$

where $u_{st}^{(2)}$ are explicit pairwise interaction modes.

Track, jointly, the conformal fraction $F_{\perp}(V)$, the tidal remainder ratio $G_{\perp}(V)$, and the integrated tensor-error fraction

$$T_{\perp}(V) \equiv \frac{\int_{\Omega} \|\Delta E_{\perp}(V)\|_F^2 dV}{\int_{\Omega} \|\Delta E^{\text{comp}}\|_F^2 dV}, \quad (15)$$

which answers the percentage question that $G_{\perp}$ alone cannot. The scientific quantity is the decay curve under enrichment,

$$F_{\perp}(V_1) \rightarrow F_{\perp}(V_2) \rightarrow F_{\perp}(V_3),$$

paired with the corresponding $T_{\perp}$ curve, with morphology, profile family, overlap fraction, compactness, and source count varied independently. The directional hypothesis is to be preregistered on a declared overlap measure rather than on a morphology label.

One guardrail is binding, to keep representability from becoming tautological:

No enrichment mode may be constructed from the target residual it is graded against.

Pairwise V_2 modes must be analytically or algorithmically declared before grading. Any learned V_3 modes must be trained on separate scenes and graded out-of-sample. Basis dimension, construction cost, and predictive degrees of freedom travel with $F_{\perp}$ and $T_{\perp}$ wherever they are reported. A sufficiently expressive basis can always erase a residual it was built from; such an erasure would measure nothing. Under this rule, the capacity-controlled enrichment curve – residual retained versus representation complexity – is itself a candidate result and may prove a cleaner measure of compression difficulty than any single V_n .

Kill conditions are split because the measurement and its interpretation die differently. The V_1 -relative collective curvature residual, as a measurement, cannot be killed by basis enrichment: $\Delta E_{\perp}(V_1)$ remains a valid record of V_1 's failure even if V_2 absorbs it completely. Its falsifiers are non-reproduction under resolution, implementation, registration, or contraction-convention changes. The nonseparability interpretation is what enrichment can kill: its falsifiers are absorption under V_2/V_3 , failure under profile-family variation, failure of the preregistered overlap law, or convention instability – summarily, no persistent basis-resistant remainder after controlled basis enrichment, profile-family variation, and convention variation.

Every outcome of the ladder is informative. Absorption at V_2 reveals that the object was pair-interaction structure and names its order; death under profile variation demotes it to a family artifact; erasure under convention change narrows it to an accounting choice; survival through all

three would convert nonseparability from a known qualitative difficulty into a measured, controlled quantity.

The next question is therefore not whether the fused scene is nonadditive, but how rapidly that nonadditivity becomes representable as the admissible source basis is enlarged.

Planned Figures

The numerical figure payloads are not fabricated in this draft. The publication assembly will replace the descriptions below with figures generated from certified payloads.

F1 – The construction.

Three source morphologies (clustered, filamentary, shell) rendered with matter-proxy masks and solved ψ isocontours at $f = 1$; compactness ladder schematic with the pre-declared stress rung marked.

F2 – The ablation calculus.

One scene, one source removed: fused solve, ablated solve, response field $\delta\psi_s$; inset, the screened operator effective source $\bar{\rho}_s\psi^{-3}$ against the bare $\bar{\rho}_s$.

F3 – The two-factor law.

Measured S^N/S^L against $(\mu_{\text{eff}}/\mu)C^{\text{lin}}$ across all fifteen ladder rungs; twelve in-band rungs filled and three $f = 2$ stress rungs open; residual panel with $\leq 1.6\%$ in-band medians; C^{lin} versus C_{full} inset at the stress rungs.

F4 – The codec earns its residual.

Normalized field-error distributions: raw linear twin (48–100%) versus reconstruction (0.65–1.2%) on identical axes, all nine grading scenes; the approximately $70\times$ contrast is the figure’s single message.

F5 – The two rejected location hypotheses.

ρ_{loc} values against surrogate bands; margin-affinity correlations with significance markers. Both panels marked REJECTED; opposite-tail significance explicitly labeled post-hoc.

F6 – The residual’s stable geometry.

$\delta\psi$ pattern at $f = 1$ versus $f = 2$ per morphology (correlations 0.990–0.995); 64^3 versus 96^3 pattern comparison (0.9997); one-signed exterior deficit shown in a signed rendering.

F7 – The witness gate, crossed.

Spatial rows per morphology: top, $\delta\psi_{\perp}$; middle, $\|\Delta E^{\text{comp}}\|_F$; bottom, $\|\Delta E_{\perp}\|_F$. Right margin: $G_{\perp}$ values and $f = 1 \leftrightarrow f = 2$ depth correlations. Companion panel: ψ -level squared- L^2 split, plotted in its own currency and not converted.

F8 – The destruction ladder.

Conceptual only: $V_1 \subset V_2 \subset V_3$ with the question $\Delta E_{\perp}(V_1) \rightarrow \Delta E_{\perp}(V_2) \rightarrow \Delta E_{\perp}(V_3)$? No quantitative axes.

Table T1 – Claim Register

Formal rank uses only canonical ladder rungs; “predictive” and “candidate architecture” are qualifiers, not rungs. The kill-condition column displays the falsification architecture rather than asserting rigor.

Claim	Formal rank	Qualifier / scope	Prereg.	Kill condition
Screening law $S^N/S^L \approx \mu_{\text{eff}}/\mu$, exponent -3 analytic	Established	Declared Family S construction	yes	Zero-parameter correlation failure on new rungs or independent implementations within the construction.
Two-factor response law with linearized compensation	Reproduced	Tested morphology family	yes	In-band residual blow-up under new morphologies or compactness within family.
Predictive compression (no non-linear constraint PDE solves)	Reproduced	Predictive; tested morphology family	yes	Fidelity failure under family extension; non-reproduction at higher resolution.
Compensation monotone with compactness	Observed	Tested family	yes	Non-monotone counterexample within family.
Residual is stable, resolution-reproduced geometry	Observed	Tested family	yes	Non-reproduction under resolution or implementation change.
Two-species decomposition $\delta\psi = \delta\psi_{\text{rw}} + \delta\psi_{\perp}$	Observed	Post-hoc; promotion candidate	post-hoc	Sensitivity to declared projection convention (mask, inner product, constant mode).
Collective curvature residual $\Delta E_{\perp}$ (summary statistic $G_{\perp}$)	Observed	V_1 -relative; declared convention; tested family	yes	Non-reproduction under resolution, implementation, registration, or contraction-convention change.
Commons-as-nonseparability	Observed	Candidate architecture; interpretation only	no	Absorption under V_2/V_3 ; profile-family failure; overlap-law failure; convention instability.
Matter localization of residual	Rejected	Preregistered directional hypothesis	yes	Rejected; revival requires new preregistration.
Contested-margin affinity of residual	Rejected	Preregistered directional hypothesis	yes	Rejected for this residual; margin instrument itself not falsified.
One-factor exponent morphology-universality	Rejected	Preregistered corollary	yes	Rejected; compensation covariance identified as mechanism candidate.

A Derivations

A.1 Canonical form

For conformally flat time-symmetric data $\gamma_{ij} = \psi^4 \delta_{ij}$, $K_{ij} = 0$, with physical density ρ , the Hamiltonian constraint is $\nabla^2 \psi = -2\pi\psi^5 \rho$. Family S prescribes $\bar{\rho} \equiv \psi^8 \rho$, giving Eq. (F-S). The definition and the equation travel together; the two forms must not be mixed.

A.2 ADM identity

Writing $\psi = 1 + u$ with $u \rightarrow M_{\text{ADM}}/(2r)$, integrate Eq. (F-S) over a ball and apply Gauss's theorem. The surface term at infinity yields $-2\pi M_{\text{ADM}}$, whence Eq. (ADM). The bare mass $\mu_s = \int \bar{\rho}_s dV$ is the $\psi \rightarrow 1$ limit of the same integral.

A.3 Linearized ablation

One sign convention is used everywhere: $\delta\psi \equiv \psi - \psi_{\text{abl}}$, so $\psi_{\text{abl}} = \psi - \delta\psi$ and $\delta\psi > 0$. The ablated constraint is

$$\nabla^2(\psi - \delta\psi) = -2\pi\bar{\rho}_{\text{ctx}}(\psi - \delta\psi)^{-3}.$$

Expanding

$$(\psi - \delta\psi)^{-3} = \psi^{-3} + 3\psi^{-4}\delta\psi + O(\delta\psi^2)$$

and subtracting Eq. (F-S) gives Eq. (LIN-ABL). The response statistics in Sec. 3 are magnitude ratios and are unaffected by the overall sign convention; the convention matters for signed field renderings and for the derivation itself.

A.4 Screening law

For any response functional S linear in its source at leading order, comparing Eq. (LIN-ABL) with the linear twin

$$\nabla^2 \delta\psi_L = -2\pi\bar{\rho}_s$$

gives Eq. (SCREEN) at first order, up to the compensation correction.

A.5 Compensation

The ablated scene relaxes: with source s removed, ψ decreases on the surviving profiles, raising each surviving $\mu_{\text{eff},s'}$. The ADM response to ablation is therefore not $\mu_{\text{eff},s}$ but $C_s \mu_{\text{eff},s}$ with $C_s < 1$. The linearized compensation C^{lin} is obtained by evaluating the survivors' effective-mass shift using the $\delta\psi$ of Eq. (LIN-ABL). The screening-mass overlap form and full derivation follow the supplementary memo [10].

A.6 Reconstruction operator

The reconstruction is

$$\psi_{\text{rec}} = 1 + \sum_s w_s u_{L,s}, \quad w_s = \frac{\mu_{\text{eff},s}}{\mu_s},$$

with the scalar fixed point

$$\mu_{\text{eff},s} = \int \bar{\rho}_s (\psi_{\text{rec}})^{-3} dV$$

iterated to convergence. No nonlinear PDE solve appears.

A.7 Electric-Weyl identification for the declared zero-anisotropic-stress completion

Let n^μ be the unit normal, $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$ the spatial projector, $j_i = -h_i^\mu T_{\mu\nu} n^\nu$, and $S_{ij} = h_i^\mu h_j^\nu T_{\mu\nu}$. In the standard $3 + 1$ decomposition, the electric part of Weyl may be written

$$E_{ij} = \left[{}^{(3)}R_{ij} + K K_{ij} - K_i^k K_{kj} - 4\pi S_{ij} \right]^{\text{TF}}, \quad (16)$$

up to the adopted Weyl-sign convention [11]. For $K_{ij} = 0$ this reduces to Eq. (4). Therefore any matter completion with vanishing anisotropic spatial stress, $S_{\langle ij \rangle} = 0$, yields Eq. (5) exactly. The pressureless-dust-at-rest completion is one realization, but isotropic pressure would also drop out of the trace-free sector. The Hamiltonian constraint fixes the trace ${}^{(3)}R = 16\pi\rho$ on the time-symmetric slice.

Before prose freeze, Eq. (16) and the code implementation of Eqs. (2)–(3) are to receive a line-by-line provenance check against the archived implementation and standard $3 + 1$ reference.

B Solver, convergence, and certification – structural stub

Final assembly will include: Picard/DST-I implementation details and relaxation schedule; float64 discipline and the underflow rule for payload arithmetic; algebraic-residual gate ($\leq 8.1 \times 10^{-11}$) across all 306 sweep solves; monopole boundary self-consistency; $64^3/96^3$ convergence tables; independent-implementation agreement battery; and the E_{ij} construction of Sec. 2.4 reproduced verbatim from the archived implementation, with a line-by-line correspondence between displayed expressions and code.

C Preregistrations and concordance – structural stub

Final assembly will include the verbatim preregistration documents for the morphology sweep, reconstruction, residual, and curvature-bridge experiments; a concordance table mapping every registered hypothesis to its disposition and main-text location; the original 20-surrogate protocol; and the pre-filed 1,000-surrogate publication-extension declaration, with opposite-tail significance explicitly labeled post-hoc.

D Per-scene result tables – structural stub

Final assembly will include the row-expanded grading table per scene and rung; compensation medians; residual statistics with surrogate-ensemble sizes and empirical p -values; ΔE^{comp} and $G_\perp$ per scene; and a data-availability statement referencing archived result bundles and seed-deterministic roster generators.

Binding provenance rule. Every numeral appearing in prose – residual gates, solve counts, exponent ranges, compensation decimals, timings, and reported ratios – must be tied to a specific archived JSON/manifest entry. No number survives into frozen text on prose inheritance alone.

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